Lab: Agent Detective -- Who (or What) Has Goals?

Objective

Determine what qualifies as an **agent** by analyzing a range of everyday entities, from pets to thermostats to classmates, and identifying their preferred states, sensory channels, and actions.

Prerequisites

- Completed the Systems module and lab
- Read the module on Agency and Goals

Part 1: The Agent Lineup

Goal: Classify a set of entities as agents or non-agents.

For each entity below, decide whether it qualifies as an agent under Active Inference. Remember, an agent is a system that acts to maintain preferred states (minimize surprise).

Entity	Agent? (Yes/No)	What are its preferred states?	How does it sense the world?	How does it act?
Your pet (dog, cat, fish)				
A thermostat				
A rock				
A plant on a windowsill				
A self-driving car				
A classmate studying for a test				

Write your response here

Part 2: You as an Agent -- A Day in Your Life

Goal: Track your own agency over one hour.

Pick any hour of your day. Every 10 minutes, record:

1. What was your **preferred state** (what did you want)?
2. What **sensory information** did you receive?
3. What **action** did you take?
4. Did your action reduce surprise (move you closer to your preferred state)?

Time	Preferred State	Sensory Input	Action Taken	Surprise Reduced?
0 min				
10 min				
20 min				
30 min				
40 min				
50 min				

Write your response here

Part 3: Comparing Agents

Goal: Compare a biological agent to a mechanical agent.

Choose one biological agent (e.g., yourself, your pet, a bird outside) and one mechanical agent (e.g., a Roomba, a thermostat, an automatic door). Answer:

1. What do both agents have in common?
2. What can the biological agent do that the mechanical agent cannot?
3. Which agent has a more complex generative model? Why?
4. Could the mechanical agent ever be "surprised" in the same way you are? Explain.

Write your response here

Part 4: Designing an Agent

Goal: Design a simple agent on paper for a specific task.

Imagine you are designing an agent whose job is to keep a classroom comfortable. Specify:

1. **Preferred states**: What conditions does it try to maintain?
2. **Sensory inputs**: What does it measure?
3. **Available actions**: What can it do?
4. **Generative model**: What does it "believe" about how its actions affect the environment?

Write your response here

Summary Table

Concept	Definition	Your Example
Agent	A system that acts to maintain preferred states	
Preferred States	The conditions the agent "wants" to be in	
Surprise	A state the agent did not predict and does not prefer	
Generative Model	The agent's internal representation of how the world works	